

SLIMANE DRIDI, PHD

Research Scientist | Mathematical & Statistical Modelling | Python

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slimaned.github.io

Skills

Statistical inference | Stochastic processes | Simulation | Time series | Experimental design | Model validation | Game theory

Python | NumPy | SciPy | pandas | Matplotlib | pytest | Jupyter | Git | GitHub Actions | Python packaging (PyPI) | R | SQL (basics) | C++ | Mathematica | AI coding agents (daily use)

Languages

French: Native

English: Fluent

Arabic: Intermediate

Spanish: Basic conversational proficiency

Selected work

- Designed, built and maintain pyNamo-EGT, an open-source Python library for equilibrium analysis and visualisation of evolutionary game dynamics: published on PyPI with automated CI/CD releases (GitHub Actions, signed provenance), test suite, documentation and browser-based tutorials; archived with a Zenodo DOI (GitHub, PyPI).

Profile

Research scientist and mathematical modeller with 10+ years of experience building, simulating and validating quantitative models in France, Switzerland and the United States. My research focused on how agents learn and respond to incentives (reinforcement learning, evolutionary game theory), and I apply the same tools to real data: statistical modelling, simulation and model validation in Python, shipped as tested, documented and reusable software. I want to bring this modelling rigour to real-world decision and performance problems in industry.

Professional Experience

- Independent Quantitative Research, Financial Markets | Self-employed

09/2023–12/2025 — Toulouse, France

- Built Python tools for market data analysis, statistical inference and strategy evaluation.
- Designed and backtested systematic trading strategies (pairs mean reversion) on historical data, with a focus on robustness and out-of-sample performance.
- Developed an algorithmic trading system and assessed its limits under real-world conditions.

- Mathematics Instructor, Nursing Entrance Examination | French Red Cross

12/2024–01/2026 — Toulouse, France

- Mathematics, Statistics & Programming Tutor | Acadomia

09/2023–06/2026 — Toulouse, France

- Researcher | Toulouse School of Economics

04/2021–08/2021 — Toulouse, France

- Research Fellow | Institute for Advanced Study in Toulouse

09/2017–07/2020 — Toulouse, France

- Developed and analysed stochastic models of reinforcement learning and cooperation in repeated games; checked analytical predictions against simulations.

- Postdoctoral Researcher | University of Pennsylvania

08/2015–08/2017 — Philadelphia, PA, USA

- Modelled the evolution of learning in mathematical biology.

- Postdoctoral Researcher | University of Lausanne

03/2014–02/2015 — Lausanne, Switzerland

Education

- PhD in Evolutionary Game Theory | University of Lausanne

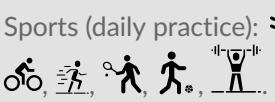
2009–2013 — Lausanne, Switzerland

- Master's Degree in Cognitive Science | University of Grenoble

09/2007–09/2009 — Grenoble, France

- Built a validated pressing-aggression index from event and player-tracking data, testing an evolutionary game theory model against 1,500+ matches across Europe's top soccer leagues (GitHub).
- Co-authored seven scientific articles (6 peer-reviewed), presented at ten conferences and seminars, and reviewed manuscripts for ten international journals (Portfolio).

Interests

- Sports (daily practice): 
- Mountain hiking.
- Volunteering: outreach work supporting people experiencing homelessness.

• Bachelor's Degree in Psychology | University of Grenoble

09/2004–09/2007 Grenoble, France

Teaching Experience

• Research Project Supervision | Toulouse School of Economics, University of Pennsylvania, University of Lausanne

2013–2020

- Supervised three Master's projects and one undergraduate research project.

• Teaching Assistant | Toulouse School of Economics, University of Lausanne

2013–2018

- Linear algebra, population genetics, and experimental design.